PROJECT 2

ANNA JOHNSON FLOORING

DSE 6100 · Fall 2026

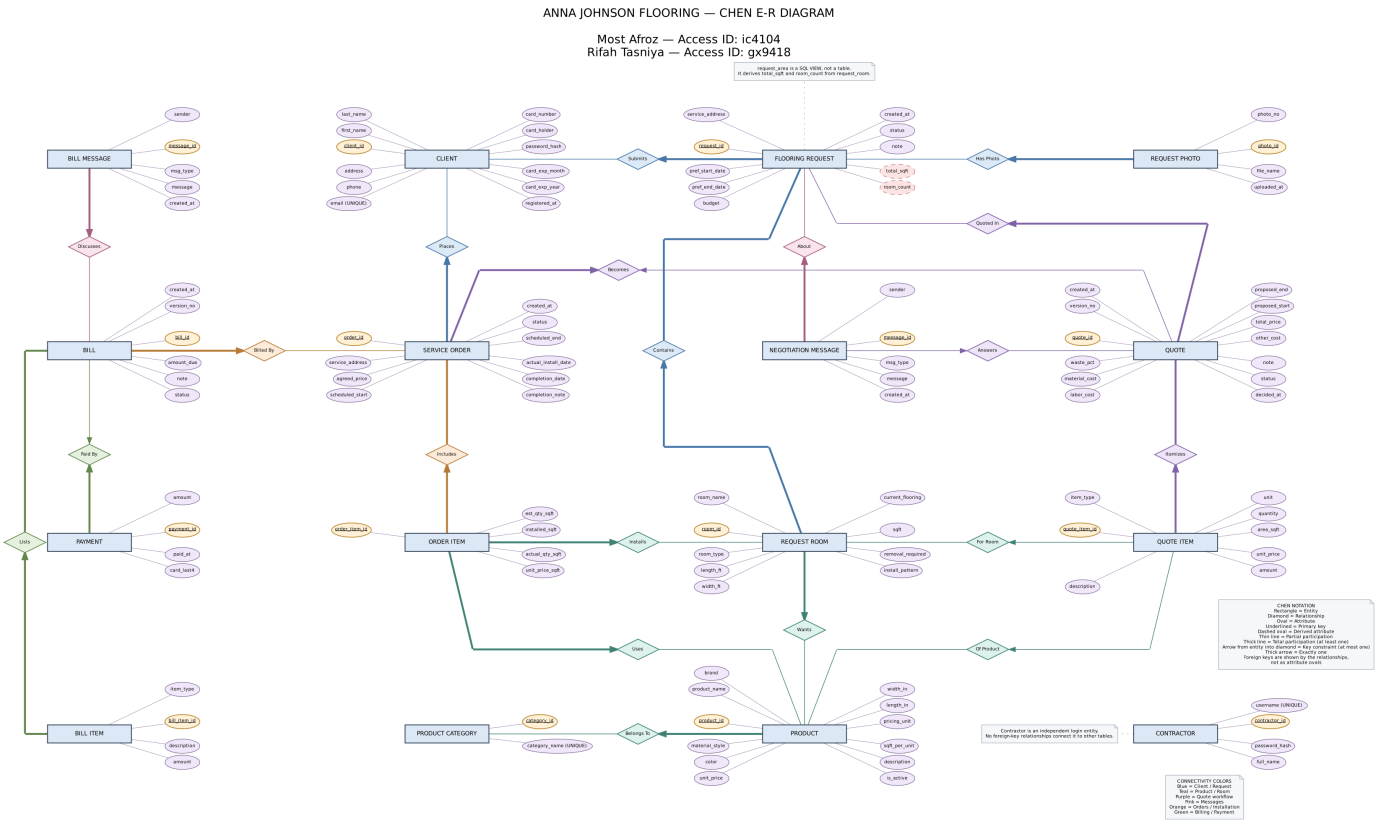
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Presentation video

https://PASTE-YOUR-VIDEO-LINK-HERE

2. E-R Diagram

The diagram uses Chen notation: rectangles represent entities, diamonds represent relationships, and ovals represent attributes. Key attributes are underlined. An arrow from an entity into a relationship marks a key constraint (the entity takes part at most once), a thick line marks total participation (at least once), and a thick arrow means exactly once. Foreign keys are represented by the relationships, so they are not drawn as attribute ovals.



3. Entities and Relationships

The following descriptions summarize the purpose of each entity and the main constraints governing how the records are related.

Entity Definitions

Contractor	Stores Anna's contractor account and login information. The project models one contractor and does not introduce separate employee roles.
Client	Stores client identity, contact information, login credentials, and the payment-card fields included for the project.
Product Category	Groups flooring products into categories.
Product	Stores flooring product details, including category, brand, material style, color, dimensions, pricing unit, and availability.
Flooring Request	Records a client's flooring project request, service address, preferred dates, budget, notes, status, and creation date.
Request Room	Represents an individual room or area included in a flooring request, including its dimensions, selected product, and installation requirements.
Request Photo	Stores uploaded photo information associated with a flooring request.
Quote	Stores a contractor's proposed price and terms for a request. Multiple versions can be retained.
Quote Item	Stores individual quote components, such as material, labor, removal, delivery, or discounts.
Negotiation Message	Preserves messages exchanged during the request and quote process.
Service Order	Records the accepted project, including its agreed price, service address, schedule, status, and completion information.
Order Item	Records the rooms and flooring products associated with a service order, including installed area and material quantities.
Bill	Stores an amount due for a service order. Different bill versions can be retained.
Bill Item	Stores individual charges that make up a bill.
Bill Message	Preserves messages and explanations associated with a bill, including disputes.
Payment	Records a payment associated with a bill, including its amount, date, and card's last four digits.
request_area (View)	Calculates request-level room-area information. It is a derived view rather than a separately stored base table.

Relationship Constraints

Client and Flooring Request	One client can submit multiple flooring requests. Each request belongs to one client through its client_id foreign key.
Product Category and Product	One product category can contain multiple products. Each product references its category.
Flooring Request and Request Room	A request can contain multiple rooms. Each room belongs to one request. Room names are unique within a request, although the same room name can appear in different requests.

Flooring Request and Request Photo	A request can have multiple photos. Photos are numbered from 1 through 5, and each request/photo-number combination is unique.
Product and Request Room	A product can be selected for multiple rooms. Each room references one desired flooring product.
Flooring Request and Quote	A request can have multiple quote versions. Each quote belongs to a request, and its version number is unique within that request.
Quote and Quote Item	A quote can contain multiple line items. Each quote item belongs to a quote and can identify the relevant room or product.
Request, Quote, and Negotiation Message	A request can have multiple negotiation messages. Messages preserve the communication history and can be associated with a quote.
Quote and Service Order	A service order references a quote. The unique quote_id constraint allows at most one service order to be created from a particular quote.
Service Order and Order Item	A service order can contain multiple order items. Each order item records the room and product associated with the installed work.
Service Order and Bill	A service order can have multiple bill versions. The combination of order_id and version_no prevents duplicate bill versions for the same order.
Bill and Bill Item	A bill can contain multiple line items, allowing its charges to be explained individually.
Bill and Bill Message	A bill can have multiple associated messages, preserving communication about the bill or a dispute.
Bill and Payment	A bill can have at most one payment record because bill_id is unique in the payment table. Each payment references one bill.

4. Relational Schema

The relations below are generated from the project's db/schema.sql file. PK identifies primary-key attributes; FK identifies foreign-key attributes and their referenced tables. The request_area view is identified separately because it is derived rather than a base table.

contractor

```
contractor_id: INT [PK], username: VARCHAR(50), password_hash: VARCHAR(255), full_name: VARCHAR(100)
```

client

```
client_id: INT [PK], first_name: VARCHAR(50), last_name: VARCHAR(50), address: VARCHAR(255), phone: VARCHAR(20),
email: VARCHAR(100), password_hash: VARCHAR(255), card_holder: VARCHAR(100), card_number: CHAR(16), card_exp_month:
TINYINT, card_exp_year: SMALLINT, registered_at: DATETIME
```

product_category

```
category_id: INT [PK], category_name: VARCHAR(50)
```

product

```
product_id: INT [PK], category_id: INT [FK to product_category], product_name: VARCHAR(100), brand: VARCHAR(100),
material_style: VARCHAR(100), color: VARCHAR(50), unit_price: DECIMAL(10,2), pricing_unit:
ENUM('sq_ft','box','piece'), length_in: DECIMAL(6,2), width_in: DECIMAL(6,2), sqft_per_unit: DECIMAL(8,2),
description: TEXT, is_active: BOOLEAN
```

flooring_request

```
request_id: INT [PK], client_id: INT [FK to client], service_address: VARCHAR(255), pref_start_date: DATE,
pref_end_date: DATE, budget: DECIMAL(10,2), note: TEXT, status:
ENUM('pending','quoted','negotiating','accepted','rejected','cancelled'), created_at: DATETIME
```

request_room

```
room_id: INT [PK], request_id: INT [FK to flooring_request], room_name: VARCHAR(50), room_type: ENUM('living
room','kitchen','bedroom','bathroom','hallway','basement','stairs','other'), length_ft: DECIMAL(7,2), width_ft:
DECIMAL(7,2), sqft: DECIMAL(9,2), product_id: INT [FK to product], current_flooring: VARCHAR(50), removal_required:
BOOLEAN, install_pattern: ENUM('straight','diagonal','herringbone','chevron','brick','other')
```

request_photo

```
photo_id: INT [PK], request_id: INT [FK to flooring_request], photo_no: TINYINT, file_name: VARCHAR(255),
uploaded_at: DATETIME
```

quote

```
quote_id: INT [PK], request_id: INT [FK to flooring_request], version_no: INT, created_at: DATETIME, waste_pct:
DECIMAL(5,2), material_cost: DECIMAL(10,2), labor_cost: DECIMAL(10,2), other_cost: DECIMAL(10,2), total_price:
DECIMAL(10,2), proposed_start: DATE, proposed_end: DATE, note: TEXT, status:
ENUM('open','superseded','accepted','rejected','cancelled'), decided_at: DATETIME
```

quote_item

```
quote_item_id: INT [PK], quote_id: INT [FK to quote], item_type:
ENUM('material','removal','labor','underlayment','trim','stairs','delivery','other','discount'), room_id: INT [FK to
request_room], product_id: INT [FK to product], description: VARCHAR(255), area_sqft: DECIMAL(9,2), quantity:
DECIMAL(10,2), unit: VARCHAR(20), unit_price: DECIMAL(10,2), amount: DECIMAL(10,2)
```

negotiation_message

```
message_id: INT [PK], request_id: INT [FK to flooring_request], quote_id: INT [FK to quote], sender:
ENUM('client','contractor'), msg_type: ENUM('counter','note','accept','reject','cancel'), message: TEXT, created_at:
DATETIME
```

service_order

```
order_id: INT [PK], quote_id: INT [FK to quote], client_id: INT [FK to client], service_address: VARCHAR(255),
agreed_price: DECIMAL(10,2), scheduled_start: DATE, scheduled_end: DATE, status:
```

ENUM('scheduled','in_progress','completed','cancelled'), created_at: DATETIME, actual_install_date: DATE,completion_date: DATE, completion_note: TEXT

order_item

order_item_id: INT [PK], order_id: INT [FK to service_order], room_id: INT [FK to request_room], product_id: INT [FK to product], installed_sqft: DECIMAL(9,2), est_qty_sqft: DECIMAL(10,2), actual_qty_sqft: DECIMAL(10,2), unit_price_sqft: DECIMAL(10,2)

bill

bill_id: INT [PK], order_id: INT [FK to service_order], version_no: INT, created_at: DATETIME, amount_due: DECIMAL(10,2), note: TEXT, status: ENUM('unpaid','disputed','superseded','paid')

bill_item

bill_item_id: INT [PK], bill_id: INT [FK to bill], item_type: ENUM('quoted','extra_material','subfloor_repair','extra_labor','upgrade','discount','credit','other'), description: VARCHAR(255), amount: DECIMAL(10,2)

bill_message

message_id: INT [PK], bill_id: INT [FK to bill], sender: ENUM('client','contractor'), msg_type: ENUM('dispute','explanation','revision','note'), message: TEXT, created_at: DATETIME

payment

payment_id: INT [PK], bill_id: INT [FK to bill], amount: DECIMAL(10,2), paid_at: DATETIME, card_last4: CHAR(4)

request_area (VIEW)

Derived room-area information calculated from the request-room data. This is a SQL view, not a separate base table.

5. Assumptions and Design Decisions

The following assumptions explain the business rules behind the database and distinguish database-enforced constraints from rules managed by the website.

1. Clients, Requests, and Rooms

- A client is identified by an automatically generated client_id. Email addresses are unique because email is used for login.
- A flooring request must contain at least one room as a business rule. Each room belongs to exactly one request. Room names are unique within a request, so two rooms in different requests can both be named Kitchen.
- Each room has one desired flooring product. If a client wants different flooring products in separate areas of the same physical space, those areas are entered separately.
- Clients can provide room length and width or enter the total square footage directly. Stairs can also be entered using their total square footage.

2. Photos, Quotes, and Negotiation

- Request photos are optional, with a maximum of five per request. The photo number range and unique combination of request_id and photo_no enforce this limit in the database.
- Anna can respond to a request with a full quote, a note, or a rejection. Clients can accept, reject, cancel, or submit a counterproposal. Counterproposals are stored as free text because proposed changes can involve different terms, such as product choice, price, or installation date.
- Quotes are versioned rather than overwritten. The combination of request_id and version_no is unique. Only the newest quote is intended to remain open; older versions become superseded. The rule about which quote is currently open is handled by the application workflow.
- A quote can contain multiple line items so material, removal, labor, underlayment, trim, stairs, delivery, and discounts can be explained separately. Quote-level material_cost, labor_cost, other_cost, and total_price provide a summary of those costs.

3. Orders, Bills, and Payments

- Accepting a quote creates a service order. The unique quote_id constraint prevents more than one service order from referencing the same quote.
- The service order preserves the agreed service address and price information. These snapshots protect the historical agreement if the client's address or a product's current price changes later.
- Bills are versioned in the same way as quotes. Each order can have multiple bill versions, identified by a unique order_id and version_no combination. The application manages which version is currently active.
- The intended payment rule is one full payment per bill version, with no partial payments. The unique bill_id constraint allows at most one payment record per bill. The application must also validate that the payment amount matches the amount due.
- A disputed bill remains unpaid. It therefore continues to qualify for the overdue rules until it is paid.

4. Reporting Rules

- **Overdue bills:** A bill is overdue when it has not been paid within seven days of its bill date. Superseded bill versions are excluded; the calculation uses the version the client is currently expected to pay.
- **Problem customers:** A problem customer has at least two overdue bills, with none of those overdue bills paid within seven days after becoming overdue. This is 14 days from the original bill date.
- **Excellent customers:** Every paid bill must have been paid within 24 hours. A bill less than 24 hours old does not disqualify a customer simply because it has not yet been paid. At least one paid bill is required.

- **Best customers:** Rankings use money actually received from the payment table, rather than quoted prices. Only completed orders are included, and the report displays the top ten customers.
- **Installed square footage:** Queries 5 and 6 use the floor area of rooms in completed orders. Material quantity is different because it includes the waste allowance.
- **Total project price:** Query 5 uses the latest valid bill, or the agreed project price when a bill has not yet been created.

5. Database and Implementation Decisions

- **Historical records:** Quotes, bills, and messages are preserved instead of being overwritten. New quote and bill versions are stored as new records, while the status of an older version can be updated. Messages are stored separately with timestamps to preserve communication history.
- **Waste allowance:** Estimated material quantity is calculated as area multiplied by (1 + waste percentage). For example, a 500-square-foot room with 10% waste requires an estimated 550 square feet of material.
- **Derived request area:** Total requested square footage is calculated through the request_area view rather than stored as a separate request-level value. This avoids maintaining a duplicate total that could become inconsistent with the individual rooms.
- **Product records:** Products referenced by historical quotes are not deleted. Products can instead be marked inactive, preserving their historical references.
- **Referential integrity:** Primary keys identify records, foreign keys connect related records, and the schema uses constraints such as NOT NULL, UNIQUE, CHECK, and referential actions where defined. Some workflow rules, such as requiring at least one room or allowing only one active quote version, must also be checked by the application.
- **Payment-card information:** Card-related fields are included for the project demonstration. A production system should use a payment processor and securely retain a token and limited card details, such as the last four digits, rather than storing a full card number.

Real-World Flooring Observation

Flooring materials are sometimes priced by square foot but sold in whole boxes. For example, if one box covers 23.64 square feet, a project requiring 550 square feet needs 24 boxes rather than 23.3 boxes. This is why the product design includes pricing_unit and sqft_per_unit, allowing box quantities to be rounded up to whole units. A 10% waste allowance is used as the default, while the contractor can adjust it for installation patterns that require more material, such as herringbone or diagonal tile.